THE MAXIMAL SEPARABLE SUBEXTENSION AND ITS PURELY INSEPARABLE QUOTIENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Standard separable-purely-inseparable factorization of an algebraic extension.

Editorial note. The label lemma-separable-first names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Fields, Lemma lemma-separable-elements: the elements of an algebraic extension that are separable over the base form an intermediate field
- Fields, Lemma lemma-irreducible-polynomials: for an algebraic element in positive characteristic there is a p-power q such that the minimal polynomial has the form P(x^q) with P separable; equivalently a suitable q-th power of the element is separable over the base
- Purely inseparable extension: an algebraic extension in which every element has a minimal polynomial of the form x minus a q-th power of an element of the base, for some p-power q

Source reference. Original reference 09H0: sources/stacks-fields/pinned-original-source.tex, source lines 1348–1348. Prerequisite pointer invoked at 1939; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09HC: sources/stacks-fields/pinned-original-source.tex, source lines 1652–1652. Prerequisite pointer invoked at 1934; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let E/F be an algebraic field extension. There exists a unique subextension $E/E_{sep}/F$ such that E_{sep}/F is separable and E/E_{sep} is purely inseparable.*

Proof. If the characteristic is zero we set $E_{sep} = E$. Assume the characteristic is $p > 0$. Let E_{sep} be the set of elements of E which are separable over F. This is a subextension by Lemma 09HC and of course E_{sep} is separable over F. Given an α in E there exists a p-power q such that α^q is separable over F. Namely, q is that power of p such that the minimal polynomial of α is of the form $P(x^q)$ with P separable algebraic, see Lemma 09H0. Hence E/E_{sep} is purely inseparable. Uniqueness is clear. □

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